

AI TRAINING – DAY 5 ASSIGNMENT

Python Loops: Repeat → Control → Break → Continue → Nest

Learning Guidance + Practical Problems + Student Deliverable

Level	Beginner → Medium
Suggested Time	3–4 hours
Core Topics	for, while, range(), break, continue, nested loops
Daily Deliverable	Loop Practice Notebook
Main Focus	Learn to think using repetition and loop control

Important: What Is the Student Deliverable?

This assignment is your learning guidance. You do not submit a separate file for every activity. Your final submission for Day 5 is ONE notebook called the Loop Practice Notebook. Put your practice, 10 problems, debugging work, real-world challenge and reflection inside that notebook.

Recommended filename: `Day_05_Loop_Practice_YourName.ipynb`

Learning Goals

- Understand for and while loops.
- Explain loops, range(), break, continue and nested loops in your own words.
- Use range() correctly.
- Understand when to use break and continue.
- Build nested loops.
- Identify common loop errors and improve your solution.
- Solve problems using loops instead of repeating code manually.
- Solve 10 loop problems including prime, palindrome and patterns.

How to Use This Assignment

1. Read and understand the concept before coding.
2. Predict output before running small examples.
3. Write your own solution before asking AI for help.
4. Run code with different inputs.
5. When code fails, identify and explain the error instead of only replacing the code.
6. Record final work in the Loop Practice Notebook.

PART A – Learn Before Coding

For every concept, explain it in your own words. Do not copy a definition.

1. for loop

When do I use a for loop?

A for loop is used when we want to repeat the block of code for n times.

Write 2 simple examples.

Ex-1: print 0-9 numbers

```
for i in range(10):  
    print(i)
```

Ex-2: print "deva" for 10 times.

```
for i in range(10):  
    print("deva")
```

2. while loop

When do I use a while loop instead of a for loop?

A while loop is useful when we need to repeat the block of code until the give condition is false.

Write 2 simple examples.

```
Ex-1: num=1  
  
while num<=10:  
    print(num)  
    num+=1
```

Ex-2: print even numbers from 1-10

```
num=2  
while num<=10:  
    print(num)  
    num+=2
```

3. range()

Experiment with:

```
range(5)
```

```
for i in range(5):  
    print(i)  
output:
```

0

1

2

3

```
4
range(1, 6)
```

```
for i in range(1,6):
    print(i)
```

output:

1

2

3

4

5

```
range(2, 11, 2)
```

```
for i in range(2,11,2):
    print(i)
```

output:

2

4

6

8

10

```
range(10, 0, -1)
```

```
for i in range(10,0,-1):
    print(i)
```

output:

10

9

8

7

6

5

4

3

2

1

Predict the values before running. Understand:

```
range(start, stop, step)
```

The stop value is normally not included.

PART B – Loop Control

1. break

Create a loop that stops when a particular number is reached. Explain what happens when break executes.

Ans:

```
for i in range(10,0,-1):  
    print(i)  
    break
```

before break it will print 10 to 1 .but now it will print only 10.

When we use break it will stop the iteration when the break is appeared .it does not iterate the remaining loop.

2. continue

Create a loop that prints 1–20 but skips multiples of 3. Explain whether continue stops the entire loop.

Ans:

```
for i in range(10,0,-1):  
    if i==3:  
        continue  
    print(i)
```

it will skip that iteration and continue the remaining iterations and print all values except that value.

3. Nested Loops

Create:

```
1 2 3  
1 2 3  
1 2 3
```

Identify the outer loop, inner loop and total inner-loop executions.

```
for i in range(3):  
    for j in range(1, 4):  
        print(j, end=" ")  
    print()
```

outer loop prints the 3 rows and inner loop manage to print values 1 2 3 .and end="" print values in same line .print() update to next for every iteration.

PART C – Predict Before You Run

Predict first. Then run and compare.

```
for i in range(5):  
    print(i)
```

ans:

it will print values from 0 and ends with 4 .because it will exclude the last value.

```
for i in range(1, 6):  
    if i == 3:  
        break  
    print(i)
```

ans:

it will print starts with 1 and stops at 2 because we break when the I became 3 .then the iteration stops.

```
for i in range(1, 6):  
    if i == 3:  
        continue  
    print(i)
```

ans:

it will print from 1 to 5.and it also skips the value 3.and continue the remaining iterations.

```
i = 1  
while i <= 5:  
    print(i)  
    i += 1
```

ans:it will print 1 to 5 values .

PART D – Solve 10 Loop Problems

For every problem record: Problem → My Approach → Code → Test Cases → Output → What I Learned.

Problem 1 – Multiplication Table

Ask for a number and print its table from 1 to 10. Focus: for + range().

```
num=int(input("enter a number:"))  
for i in range(1,11):  
    print(f"{num} * {i} ={num*i}")
```

output:

enter a number:5

5 * 1 =5

5 * 2 =10

5 * 3 =15

5 * 4 =20

5 * 5 =25

5 * 6 =30

5 * 7 =35

5 * 8 =40

5 * 9 =45

5 * 10 =50

Problem 2 – Sum of Numbers

Ask for n and calculate 1 + 2 + ... + n. Focus: accumulator + for.

```
n=int(input("enter a number:"))
sum=0
for i in range(n+1):
    sum+=i
print(sum)
```

output:

enter a number:6

21

Problem 3 – Count Even and Odd Numbers

For 1 to n, count even and odd numbers. Focus: for + if.

```
n=int(input("enter a number:"))
even=0
odd=0
for i in range(1,n+1):
    if i %2==0:
        even+=1
    else:
        odd+=1
print(f"the odd numbers are : {odd}")
print(f"the even numbers are : {even}")
```

output:

enter a number:6

the odd numbers are : 3

the even numbers are : 3

Problem 4 – Prime Number Checker

Determine whether a number is prime. First explain what makes a number prime.

Ans:

If a numbers is said to prime if it divisible by only 1 and itself.

```
num=int(input("enter number:"))
count=0
for i in range(1,num+1):
    if num%i==0:
```

```

        count+=1
    else:
        count+=0
if count==1:
    print(f"{num} is prime")
else:
    print(f"{num} is not a prime")

```

output:

enter number:9

9 is not a prime

Problem 5 – Palindrome Number

Check whether a number such as 121 is a palindrome. Try without a ready-made palindrome function.

Ans: a number is said to be palindrome if it equals to reverse of the same number.

```

num=int(input("enter a number:"))
reverse=0
while num>0:
    number=num%10
    reverse=reverse*10+number
    num=num//10
if reverse==num:
    print("it is a plindrome")
else:
    print("it is not a palindrome")

```

test case1:

enter a number:121

it is a palindrome

test case 2:

enter a number:123

it is not a palindrome

Problem 6 – Find Factors

Print all factors of a number. Example: 12 → 1, 2, 3, 4, 6, 12.

```

num=int(input("enter number:"))
for i in range(1,num+1):
    if num%i==0:
        print(i,end=" ")

```

test case ;

enter number:12

1 2 3 4 6 12

Problem 7 – break Challenge

Search 1–100 and stop at the first number divisible by both 7 and 9. Use break.

```
for i in range(1,100):  
    if i%7==0 and i%9==0:  
        print(i)  
        break
```

output: 63

Problem 8 – continue Challenge

Print 1–50 but skip numbers divisible by 3. Use continue.

```
for i in range(1,50):  
    if i%3==0 :  
        continue  
    print(i)
```

output:

1,2,4,5,7,8,10,11,13,14,16,17,19,20,22,23,25,26,28,29,31,32,34,35,37,38,40,41,43,44,46,47,49.

Problem 9 – Pattern Printing

Print *, **, ***, ****, ***** and the reverse. Bonus: centered pyramid.

```
for i in range(6):  
    print(i*" ",end=" ")  
    print()  
    i+=1
```

output:

*

**

reversed:

```
for i in range(5,0,-1):  
    print(i*" ",end=" ")  
    print()
```

output:

**

*

Problem 10 – Number Pattern

Print 1, 12, 123, 1234, 12345 and then 1, 22, 333, 4444, 55555. Use nested loops.

```
for i in range(1, 6):  
    for j in range(1, i + 1):  
        print(j, end="")  
    print()
```

output:

1
12
123
1234
12345

```
for i in range(1, 6):  
    for j in range(i):  
        print(i, end="")  
    print()
```

output:

1
22
333
4444
55555

PART E – Debugging Lab

For every bug: identify the problem, explain why it happens, fix it and test it.

Bug 1
for i in range(5)
 print(i)

why it happens: it is because of missing colon(:) at the end of for loop line.

fix it:

```
for i in range(5):  
    print(i)
```

test it: executed successfully.

What is missing?

```
Bug 2
i = 1
while i <= 5:
    print(i)
```

why it happens: without giving the incrementation it does not satisfy the condition ($i \leq 5$).

fix it:

```
i = 1
while i <= 5:
    print(i)
    i+=1
```

test it:executed successfully.

Why can this become an infinite loop? Fix it.

```
Bug 3
for i in range(1, 10):
    if i == 5:
        break
    print(i)
    print('Loop completed')
```

why it happens: loop actually stopped before 5.means it will print upto 4.and also the last print statement should be outside the loop.

```
for i in range(1, 10):
    if i == 5:
        break
    print(i)
print('Loop completed')
```

test it:executed successfully.

Understand where the loop actually stops.

```
Bug 4
for i in range(1, 10):
    if i % 2 == 0:
        continue
    print(i)
```

why it happens: it is because we gave print statement after the continue so it will skip when ever the condition is correct.

fix it:

```
for i in range(1, 10):
    if i % 2 == 0:
        continue
    print(i)
```

test it:executed successfully.

Why will print(i) never execute for even numbers?

```
Bug 5
for i in range(3):
    for j in range(3):
        print('*')
```

Predict how many times * prints, then verify.

Ans: it will print 9 times because second loop is depend on first loop. For 1 iterate of first loop the second loop iterates 3 times.

PART F – Real-World Challenge

Choose ONE: ATM transaction loop; Login system with 3 attempts; Shopping cart total calculator; Student marks analyzer; Number guessing game.

- Use a loop.
- Use at least one if.
- Use at least one break OR continue.
- Take user input.
- Display proper output.
- Test at least 3 different cases.

Number guessing game:

```
secret_number = 7
while True:
    guess = int(input("Guess a number between 1 and 10: "))
    if guess == secret_number:
        print("Correct! You guessed the number.")
        break
    elif guess > secret_number:
        print("Too high! Try again.")
    else:
        print("Too low! Try again.")
```

output:

Guess a number between 1 and 10: 5
Too low! Try again.

Guess a number between 1 and 10: -1
Too low! Try again.

Guess a number between 1 and 10: -1
Too low! Try again.

Guess a number between 1 and 10: 5
Too low! Try again.

Guess a number between 1 and 10: 6
Too low! Try again.

Guess a number between 1 and 10: 7
Correct! You guessed the number.

PART G – Explain in Your Own Words

7. What is a loop?
Ans: loop is nothing but repeating the block of code .
8. What is the difference between for and while?
Ans: for loop is used when we know how many times to iterate the loop.while loop is used when we don't know how many times to iterate.it iterate until the given condition is wrong.
9. What does range() do?
Ans: range() is mostly used in for when we want to print values in between a number.syntax for range id (start,end) it will exclude the end value.
10. What happens when break executes?
Ans: when break executes the loop will stop iterating.
11. What happens when continue executes?
Ans: when continue executes then it will skip the iteration and continue the iterating the remaning.
12. What is a nested loop?
Ans: nested loop is nothing but a loop inside a loop .in this situation the inner loop is depends on the outer loop.
13. When can a while loop become infinite?
Ans:while loop become infinite when it condition never becomes False.
14. Why is indentation important inside loops?
Ans: the most important part of loop is indentation.with some space there may lead to error. With identation it wil tells which statement belongs to which loop.
15. What was the hardest problem today?
Ans:the hardest problem is printing patterns.
16. What mistake did you make and how did you fix it?
Ans:I make mistake by using slicing on integer values . by this I had learn tha we cannot use slicing directly on intergers.

PART H – Self Check

Skill	Comfortable	Need Practice
for loop	☒	☐
while loop	☒	☐
range()	☒	☐
break	☒	☐
continue	☒	☐
Nested loops	☐	☒
Prime number	☒	☐
Palindrome	☒	☐
Pattern printing	☐	☒
Debugging loops	☒	☐

STUDENT DELIVERABLE – WHAT YOU MUST SUBMIT

Submit ONE notebook only. Everything below should be inside your Loop Practice Notebook.

Day_05_Loop_Practice_YourName.ipynb

Your notebook must contain:

17. Concept Practice – your own examples for for, while, range(), break, continue and nested loops.
18. 10 Loop Problems – complete solutions for all 10 problems.
19. Debugging – corrected versions and explanations.
20. Real-World Challenge – one completed loop-based project.
21. Reflection – answers to the 10 explanation questions.
22. Test Evidence – at least 2–3 test cases for important programs.

You DO NOT need to submit:

- A separate file for each problem.
- A separate debugging file.
- A separate notes file.
- A separate Word or PDF document.
- Screenshots for every program.

Simple Rule

**LEARN → PRACTICE → SOLVE 10 PROBLEMS → DEBUG → BUILD → REFLECT →
SUBMIT ONE NOTEBOOK**

Submission Checklist

- ☐ Concept practice completed.
- ☐ All 10 loop problems completed.
- ☐ Prime number completed.
- ☐ Palindrome completed.
- ☐ Pattern problems completed.
- ☐ Debugging completed.
- ☐ Real-world challenge completed.
- ☐ Reflection completed.
- ☐ Test cases included.
- ☐ Notebook filename follows the required format.
- ☐ Final notebook is ready to submit.

DAY 5 PRINCIPLE

Don't just repeat code. Learn to control repetition.